

Multiple Disease Prediction System using ML

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Abstract

Predictive analytics in healthcare is a difficult endeavour. High-quality and sufficient data is often scarce, making it difficult to train accurate models. Identifying the most relevant features for prediction can be complex, requiring both statistical methods and domain expertise. Additionally, medical datasets are often imbalanced, with far more healthy cases than disease cases, which can skew the model's performance.

But a predictive model can eventually assist practitioners in making timely decisions regarding patients' health and treatment based on massive data. Diseases like Breast cancer, diabetes, and heart-related diseases are causing many deaths globally but most of these deaths are due to the lack of timely check-ups of the diseases. Other diseases like Parkinson's disease, while not typically fatal, it significantly impacts quality of life and can lead to severe disability. The above problem occurs due to a lack of medical infrastructure and a low ratio of doctors to the population. In this work, breast cancer, Parkinson's disease, heart, and diabetes are included. To make this work seamless and usable by the mass public, a medical test web application is made that makes predictions about various diseases using the concept of machine learning. In this work, our aim to develop a disease-predicting web app that uses the concept of machine learning-based predictions about various diseases.

Introduction

1.1 Problem Statement

Data collection and processing of that data is a tricky and worrisome task in healthcare. In this day and age of digital era, a vast quantity of multidimensional data on patients is created. The enormous, dense, and complex data must be processed and evaluated in order to extract knowledge for effective decision making. By identifying significant patterns and detecting correlations and relationships among many variables in huge databases. Diagnosis of chronic illnesses is a vital issue in the medical industry since it is based on many symptoms. It is a complex procedure that frequently leads to incorrect assumptions. When diagnosing illnesses, the clinical judgment is based mostly on the patient's symptoms as well as the physicians' knowledge and experience.

Data mining and machine learning approaches are making significant efforts to intelligently translate accessible data into valuable information in order to improve the diagnostic process's efficiency. Several studies have been conducted to explore the use of machine learning in terms of diagnostic abilities. It was discovered that, when compared to the most experienced physician, who can diagnose with 79.97% accuracy, machine learning algorithms could identify with 91.1% correctness. Machine learning techniques are explicitly used to illness datasets to extract features for optimal illness diagnosis, prediction, prevention, and therapy. Below are the diseases we will be working with in the scope of this article.

Breast cancer is a common and potentially deadly disease affecting many individuals worldwide. Early diagnosis is crucial for effective treatment and improved survival rates. Machine learning models have become instrumental in breast cancer diagnosis, leveraging large datasets of patient records and clinical data. These models can identify patterns and anomalies with high accuracy, often surpassing traditional diagnostic methods. By analyzing various factors such as menopausal status, tumor size, patient histories, etc. machine learning can help detect cancer at early stages.

Diabetes is a widespread chronic disease that affects millions of people globally, characterized by high blood sugar levels due to the body's inability to produce or effectively use insulin. Early diagnosis and management are crucial to prevent severe complications. Machine learning is used to diagnose and manage

diabetes by analyzing large datasets of patient records and clinical data. These models can identify patterns and risk factors, such as glucose, blood pressure, insulin, etc. to predict the onset of diabetes.

Heart Disease is a leading cause of death worldwide, encompassing various conditions that affect the heart's function. Diagnosis using machine learning approach can help manage heart disease by analyzing large datasets of patient records and clinical data. These models can identify patterns and risk factors, such as age, cholesterol, biomarkers, etc. to predict the chances of heart disease.

Parkinson's disease is a progressive neurological disorder that primarily affects movement, leading to symptoms such as tremors, rigidity, and impaired coordination. While this disease shares common symptoms with several other neurological and movement disorders, which can make it challenging to predict and diagnose accurately. Symptoms like tremors, rigidity, and balance issues are not unique to Parkinson's and can overlap with those of other conditions, leading to potential misdiagnoses, so the we make predictions by training the model with a range of biomedical voice measurements.

By incorporating machine learning, healthcare providers can offer more precise and timely interventions, ultimately enhancing patient outcomes and quality of life.

1.2 Objective

The main objective of this project is to make a predictive system using the data available from that will be used to identify and diagnose diseases like breast cancer, diabetes, heart related disease and Parkinson's disease. The main goal is to create a predictive system using available data to identify and diagnose diseases such as breast cancer, diabetes, heart-related diseases and Parkinson's disease. The project will focus on developing an interactive and user-friendly platform with React, ensuring a clean, intuitive, and responsive user interface. This interface will include forms for users to input health data like age, weight, medical history, and other relevant metrics.

Appropriate machine learning algorithms will be selected for each disease, such as logistic regression, naive bayes, decision tree classifier and random forest classifier for breast cancer, logistic regression, naive bayes and SVM for diabetes detection, logistic regression and naive bayes for heart disease, and logistic regression and SVM for Parkinson's disease. These models will be trained using large datasets from medical research databases and validated for accuracy. The trained models will then be evaluated based on their accuracy scores using training dataset and testing dataset. The model with highest accuracy score for that disease will be used to build the predictive system.

The integration and testing phase will connect the React frontend to the backend API, ensuring seamless data exchange between the user interface and the machine learning models. Thorough user testing will be conducted to ensure the system's usability, accuracy, and reliability, with feedback gathered to refine and improve the user experience. Performance optimization will be carried out

to ensure the application can handle a large number of users and data inputs efficiently.

1.3 Summary

In summary, this project aims to create a powerful and accessible tool for predicting multiple diseases through a React website integrated with machine learning models. The main objective is to develop a predictive system using the data available to identify and diagnose diseases like breast cancer, diabetes, heart-related diseases and Parkinson's disease. This system will provide users with valuable insights into their health, enabling early detection and proactive management of these serious conditions. In summary, this project aims to create a powerful and accessible tool for predicting multiple diseases through a React website integrated with machine learning models. The main objective is to develop a predictive system using the data available to identify and diagnose diseases such as breast cancer, diabetes, heart-related diseases and Parkinson's disease. This system will provide users with valuable insights into their health, enabling early detection and proactive management of these serious conditions.

The React-based platform will be designed to be user-friendly and highly interactive, ensuring that users can easily navigate and input their health data. The system will utilize advanced machine learning algorithms to analyze this data and provide accurate predictions. By identifying potential health issues early, users can take timely action, seeking medical advice and interventions that can significantly improve their health outcomes.

Literature Survey

2.1 Related Work

A structural model and a collection of conditional probabilities are used by Bayesian classifiers. They make the assumption that the contributions of all factors are independent. It first calculates the prior probability for each class, and then applies the occurrence of each variable value to an unknown scenario. A Bayes network classifier is built on a Bayesian network, which reflects a joint probability distribution over a set of category characteristics.

The SVM method and the Nave Bayes technique were used to predict kidney disease. The authors attempted to categorize various stages of kidney disease using the suggested ANFIS algorithm. The study's purpose was to design an effective categorization algorithm using several assessment metrics such as accuracy and execution time. While the SVM Algorithm provided higher classification accuracy, the Nave Bayes fared better since it produced results in less time. The results show that SVM outperforms the Nave Bayes Approach in predicting renal illness.

The fuzzy technique with a membership function was used to forecast cardiac disease. Using the Fuzzy KNN Classifier, the authors attempted to eliminate ambiguity and uncertainty from data. The 550-record dataset was separated into 25 classes, with each class having 22 items. The dataset was separated into two equal parts: training and testing. The fuzzy KNN methodology was implemented after pre-processing techniques were used. This technique was examined using several assessment metrics such as accuracy, precision, and recall, among others. Based on the data, it was discovered that the fuzzy KNN classifier outperformed the KNN classifier in terms of accuracies.

For the prediction of cardiac disease, a novel technique based on the ANN algorithm was devised. The researchers created an interactive prediction method based on categorization using an artificial neural network algorithm and taking into account the thirteen most important clinical parameters. The suggested method proved effective for predicting heart disease with an accuracy of 80% and can be very useful for healthcare practitioners.

Authors in presented an automated approach for answer- ing difficult inquiries for heart disease prediction. The Naive Bayes methodology was used to create this intelligent system in order to provide quick, better, and more accu-

rate outcomes. It might aid doctors in making clinical judgments about heart attacks. This system may be enhanced by including SMS functionality, building

Android and IOS mobile applications, and including a pacemaker in the order.

Diabetes and breast cancer were diagnosed by incorporating the adaptivity characteristic into support vector machines. The goal was to offer a rapid, automated, and adaptable diagnostic method using adaptive SVM. To achieve better results, the bias value in conventional SVM was changed. The suggested classifier produced output in the form of 'if-then' rules. The proposed method was used to diagnose diabetes and breast cancer, and it provided 100% right classification rates for both conditions. Future research should focus on developing more efficient ways for changing the bias value in conventional SVM.

For the prediction of type 2 diabetes, a hybrid model based on clustering followed by classification was proposed. For prediction, the suggested model uses K-means clustering and the C4.5 classification method with k-fold cross-validation. The model generated encouraging results with a classification accuracy of 88.38% using the hybrid technique, which might be highly useful for clinicians in making appropriate clinical choices related to diabetes.

Methodologies

In this framework, machine learning algorithms- support vector machine (SVM), logistic regression, random forest classifier, xgboost classifier, decision tree classifier and Naive Bayes are used. In frontend, React is used to develop a website, which is connected to python flask backend.

3.1 Machine learning

Machine learning is a subset of artificial intelligence (AI) that involves the development of algorithms and statistical models enabling computers to perform specific tasks without explicit instructions. Instead, these systems learn from and make decisions based on data. By recognizing patterns and correlations within large datasets, machine learning models can make predictions or decisions that improve over time as they are exposed to more data. Machine learning is applied in various domains like speech recognition, medical diagnosis, financial forecasting, and autonomous systems, to create solutions that can adapt and optimize their performance autonomously.

3.2 Unsupervised Learning

Unsupervised learning is a machine learning technique where algorithms learn from unlabeled data. Unlike supervised learning, there's no predefined output or target variable. The goal is to discover hidden patterns, structures, or groupings within the data. Common methods include clustering (grouping similar data points) and dimensionality reduction (simplifying complex data).

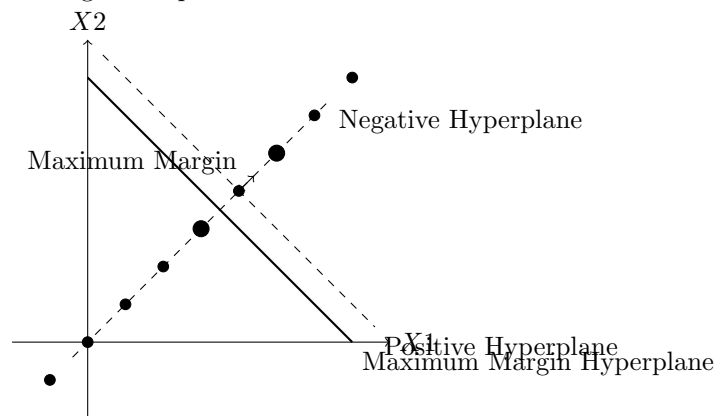
3.3 Supervised Learning

Supervised learning involves training a machine learning model on labeled data. This means the data includes both inputs and desired outputs. The model learns to map inputs to outputs, enabling it to make predictions or classifications on new, unseen data. It's like teaching a child to recognize animals by showing them pictures with labels. Common tasks include regression (predicting numerical values) and classification (categorizing data).

The biggest difference between supervised and unsupervised machine learning is the type of data used. Supervised learning uses labeled training data, and unsupervised learning does not. uses a sample dataset to train itself to make predictions, iteratively adjusting itself to minimize error. These datasets are labeled for context, providing the desired output values to enable a model to give a “correct” answer. In contrast, unsupervised learning algorithms work independently to learn the data’s inherent structure without any specific guidance or instruction. You simply provide unlabeled input data and let the algorithm identify any naturally occurring patterns in the dataset.

3.4 Support Vector Machine (SVM)

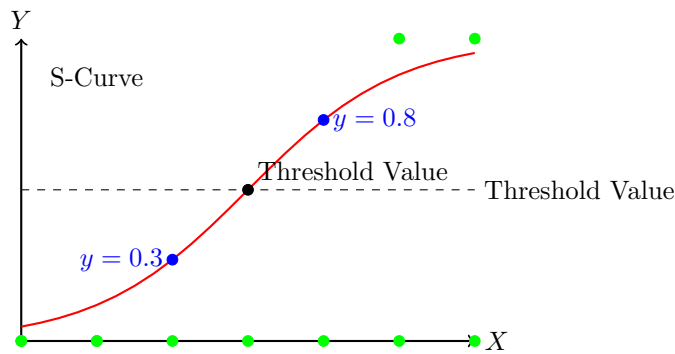
Support Vector Machines (SVMs) are a type of supervised learning algorithm that can be used for classification or regression tasks. The main idea behind SVMs is to find a hyperplane that maximally separates the different classes in the training data. This is done by finding the hyperplane that has the largest margin, which is defined as the distance between the hyperplane and the closest data points from each class. Once the hyperplane is determined, new data can be classified by determining on which side of the hyperplane it falls. SVMs are particularly useful when the data has many features, and/or when there is a clear margin of separation in the data.



3.5 Logistic Regression

Logistic regression is one of the most popular Machine Learning algorithms, which comes under the Supervised Learning technique. It is used for predicting the categorical dependent variable using a given set of independent variables. Logistic regression predicts the output of a categorical dependent variable. Therefore the outcome must be a categorical or discrete value. It can be either Yes or No, 0 or 1, true or False, etc. but instead of giving the ex-

act value as 0 and 1, it gives the probabilistic values which lie between 0 and 1. Logistic Regression is much similar to the Linear Regression except that how they are used. Linear Regression is used for solving Regression problems, whereas Logistic regression is used for solving the classification problems. In Logistic regression, instead of fitting a regression line, we fit an "S" shaped logistic function, which predicts two maximum values (0 or 1). The curve from the logistic function indicates the likelihood of something such as whether the cells are cancerous or not, a mouse is obese or not based on its weight, etc. Logistic Regression can be used to classify the observations using different types of data and can easily determine the most effective variables used for the classification. The below image is showing the logistic function.



3.6 Random Forest Classifier

Random Forest algorithm is a powerful tree learning technique in Machine Learning. It works by creating a number of Decision Trees during the training phase. Each tree is constructed using a random subset of the data set to measure a random subset of features in each partition. This randomness introduces variability among individual trees, reducing the risk of overfitting and improving overall prediction performance. In prediction, the algorithm aggregates the results of all trees, either by voting (for classification tasks) or by averaging (for regression tasks). This collaborative decision-making process, supported by multiple trees with their insights, provides an example stable and precise results. Random forests are widely used for classification and regression functions, which are known for their ability to handle complex data, reduce overfitting, and provide reliable forecasts in different environments.

Ensemble of Decision Trees: Random Forest leverages the power of ensemble learning by constructing an army of Decision Trees. These trees are like individual experts, each specializing in a particular aspect of the data. Importantly, they operate independently, minimizing the risk of the model being overly influenced by the nuances of a single tree. **Random Feature Selection:** To ensure that each decision tree in the ensemble brings a unique perspective, Random Forest employs random feature selection. During the training of each

tree, a random subset of features is chosen. This randomness ensures that each tree focuses on different aspects of the data, fostering a diverse set of predictors within the ensemble.

Bootstrap Aggregating or Bagging: The technique of bagging is a cornerstone of Random Forest’s training strategy which involves creating multiple bootstrap samples from the original dataset, allowing instances to be sampled with replacement. This results in different subsets of data for each decision tree, introducing variability in the training process and making the model more robust. **Decision Making and Voting:** When it comes to making predictions, each decision tree in the Random Forest casts its vote. For classification tasks, the final prediction is determined by the mode (most frequent prediction) across all the trees. In regression tasks, the average of the individual tree predictions is taken. This internal voting mechanism ensures a balanced and collective decision-making process.

3.7 XGBoost

XGBoost is an optimized distributed gradient boosting library designed for efficient and scalable training of machine learning models. It is an ensemble learning method that combines the predictions of multiple weak models to produce a stronger prediction. XGBoost stands for “Extreme Gradient Boosting” and it has become one of the most popular and widely used machine learning algorithms due to its ability to handle large datasets and its ability to achieve state-of-the-art performance in many machine learning tasks such as classification and regression. One of the key features of XGBoost is its efficient handling of missing values, which allows it to handle real-world data with missing values without requiring significant pre-processing. Additionally, XGBoost has built-in support for parallel processing, making it possible to train models on large datasets in a reasonable amount of time. It sequentially adds trees to correct errors from previous ones, optimizing a custom objective function with regularization to prevent overfitting. It’s known for its efficiency and speed.

3.8 Decision Tree Classifier

Decision trees are used extensively for categorizing huge datasets. Decision trees categorize data between the root node and the leaf node. The produced tree can be used for rule-making. Decision trees are rules that are easy to understand. Many decision-tree algorithms are available, including ID3, C4.5, and CART. The algorithm C4.5 for data mining is a complex decision tree approach. The idea is based on the profit ratio. The main benefits of the C4.5 algorithm are its well-functioning with both categorical and continuous features. It can also handle missing values correctly while running and utilizes less memories. It has the inconveniences of branches that are excessive and insignificant. The information gain is the basis of the ID3 algorithm. CART is a generator of

a binary decision tree, which is based on the measure of the Gini index. The ID3 algorithm has discrete features that do not manage missing values. It is a flowchart-like structure where nodes represent features, branches represent decision rules, and leaf nodes represent outcomes. It splits data into subsets based on feature homogeneity, using measures like Gini impurity or entropy. The tree is built recursively until a stopping condition is met, and predictions are made by traversing the tree from root to leaf.

3.9 Naive Bayes

Naive Bayes is a probabilistic classifier based on Bayes' theorem, assuming feature independence given the class. It calculates the posterior probability of a class given the features using prior probabilities and likelihoods. There are variations like Gaussian Naive Bayes for normally distributed features, Multinomial Naive Bayes for discrete counts, and Bernoulli Naive Bayes for binary features. Predictions are made by selecting the class with the highest posterior probability. The existence or absence of a particular class feature does not depend on the presence or absence of any other feature. It operates on the basis of conditions. If B represents the dependent event and A represents the last event the theorem Bayes may be phrased as follows: $\text{Sample (B supplied in A)} = \frac{\text{Sample (A and B)}}{\text{Sample (A)}} \cdot \text{Sample (B)}$. The approach divides the number of events in which A and B occur together by the number of circumstances in which A occurs to get the likelihood of B given A alone. In order to estimate the parameters (variable media and variances), the Naive Bayes Classifier benefits from only a few training data. Due to the assumption of independent variables, all the variances must be computed for each class. It is relevant to binary as well as multi-class problems.

3.10 React.js

React.js is a popular JavaScript library developed by Facebook for building user interfaces, particularly single-page applications. It allows developers to create large web applications that can update and render efficiently in response to data changes. React's core concept is the component-based architecture, where the UI is divided into reusable components. It is used for the frontend in machine learning projects because it promotes reusability and maintainability, enhances performance through its virtual DOM, simplifies the process of designing interactive UIs, and has a strong ecosystem and community support. This makes it ideal for creating dynamic and responsive interfaces that can handle real-time data updates, which is crucial for displaying machine learning results interactively.

3.11 Flask

Python Flask is a lightweight and flexible micro web framework written in Python. It is designed to get applications up and running quickly with a minimal amount of code, providing essential features like routing, templating, and session management without the need for heavy configuration. Flask is used for the backend in machine learning projects because of its simplicity and flexibility, seamless integration with Python's extensive machine learning libraries, excellent support for creating RESTful APIs, rich extensions for adding functionality, and its ability to accelerate the development process. Flask's straightforward approach enables quick iterations and deployment, which is essential for machine learning projects that require rapid prototyping and testing.

System Analysis

4.1 Existing System

The existing system focuses on predicting diabetes, heart disease, and Parkinson's disease using a variety of machine learning algorithms, including Naive Bayes, Decision Trees, Random Forest, SVM, and Logistic Regression. SVM achieved 76% accuracy in diabetes prediction and 71% in Parkinson's disease, highlighting its effectiveness. Other algorithms like logistic regression and Decision Trees are also employed, each leveraging distinct data characteristics for accurate predictions. Overall, the system demonstrates the potential of ML algorithms in disease prediction, with further enhancements to improve accuracy.

4.1.1 Disadvantages

- SVM achieved 76% accuracy in diabetes prediction and 71% in Parkinson's disease, which may not be sufficient for clinical use where higher accuracy is crucial.
- Other algorithms' accuracy levels are not mentioned, and they might be lower, which could affect overall reliability.
- Algorithms like SVM and Random Forest, while powerful, are often considered "black boxes," making it difficult to interpret how decisions are made, which is crucial in healthcare applications.
- While Streamlit offers a user-friendly interface, it might not provide the necessary flexibility and customization required for more complex user interactions or integration with other tools.

4.2 Proposed System

The proposed methodology for this project involves leveraging multiple training models for disease prediction, where their performances are compared. This process involves utilizing various libraries like pandas for data handling, numpy for numerical operations, scikit-learn for model training and evaluation, and pickle for exporting the trained model for future application use. Additionally,

the approach enables the simultaneous prediction of multi diseases, streamlining the prediction process for users and potentially reducing mortality rates. Compared to existing systems, this method offers faster predictions and numerous advantages, contributing to improved healthcare outcomes.

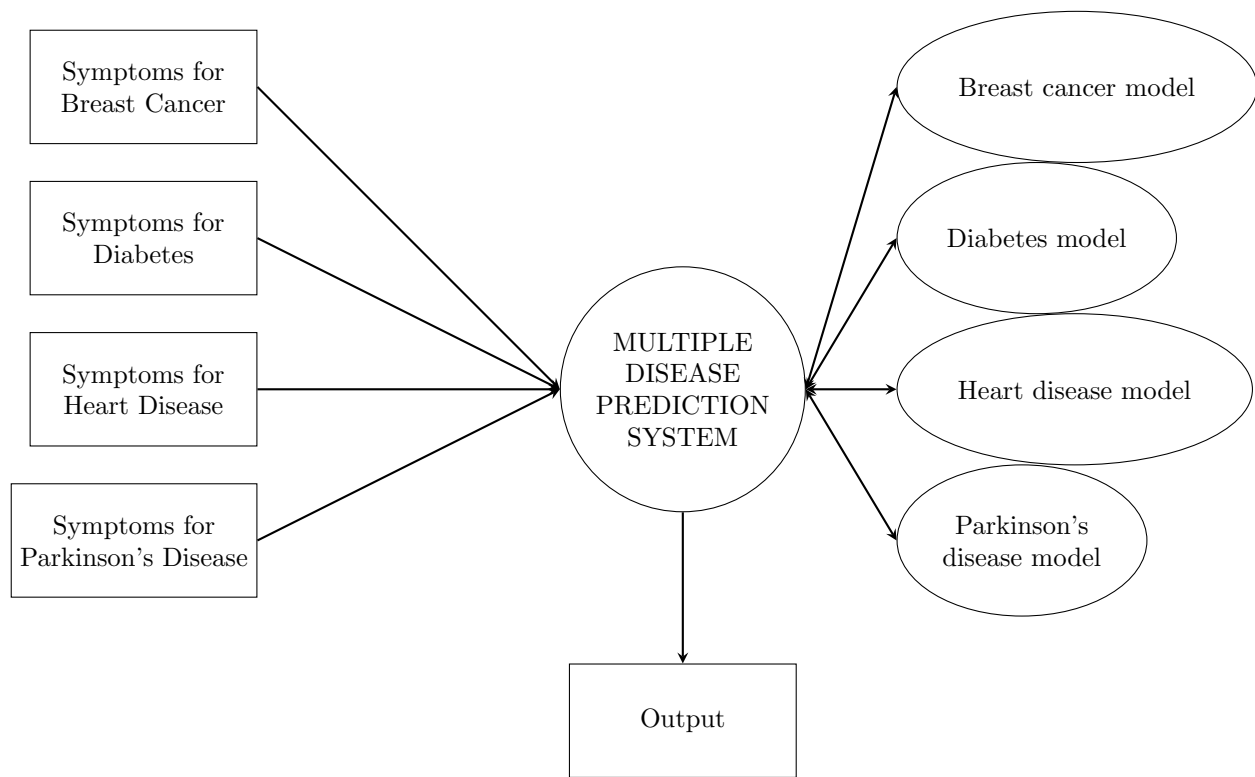
And for the front end React.js is used over Streamlit. React.js, being a highly flexible and widely adopted JavaScript library for building user interfaces, provides developers with the tools to create dynamic and responsive web applications. Unlike Streamlit, which is primarily designed for creating data-driven web apps quickly and with minimal code, React.js offers a more comprehensive and customizable approach. This flexibility is crucial for building complex interfaces where user experience and interactivity are paramount.

4.2.1 Advantages of Proposed System

- Using React.js over Streamlit for the user interface of a disease prediction system offers several significant advantages, enhancing both the functionality and user experience of the application
- Additionally, the approach enables the simultaneous prediction of multi diseases, streamlining the prediction process for users and potentially reducing mortality rates. Compared to existing systems, this method offers faster predictions and numerous advantages, contributing to improved healthcare outcomes.
- The proposed system add one more disease, Breast cancer.

4.3 Data Flow Diagram





Implementation

5.1 Dataset Collection

The datasets used to train the model are opensource.

5.1.1 Breast Cancer Dataset

The data set contains patient records from a 1984-1989 trial conducted by the German Breast Cancer Study Group (GBSG) of 720 patients with node positive breast cancer; it retains the 686 patients with complete data for the prognostic variables. These data sets are used in the paper by Royston and Altman(2013). The Rotterdam data is used to create a fitted model, and the GBSG data for validation of the model. The paper gives references for the data source.

```
RangeIndex: 686 entries, 0 to 685
Data columns (total 12 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   Unnamed: 0  686 non-null   int64
 1   pid         686 non-null   int64
 2   age         686 non-null   int64
 3   meno        686 non-null   int64
 4   size        686 non-null   int64
 5   grade       686 non-null   int64
 6   nodes       686 non-null   int64
 7   pgr         686 non-null   int64
 8   er          686 non-null   int64
 9   hormon      686 non-null   int64
10   rfstime     686 non-null   int64
11   status      686 non-null   int64
dtypes: int64(12)
memory usage: 64.4 KB
```

5.1.2 Diabetes Dataset

The dataset contains features such as number of pregnancies, glucose, blood pressure, skin thickness, insulin, body mass index (BMI) and age.

```
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Pregnancies                           768 non-null    int64
1   Glucose                               768 non-null    int64
2   BloodPressure                         768 non-null    int64
3   SkinThickness                         768 non-null    int64
4   Insulin                              768 non-null    int64
5   BMI                                   768 non-null    float64
6   DiabetesPedigreeFunction              768 non-null    float64
7   Age                                   768 non-null    int64
8   Outcome                               768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
```

5.1.3 Heart Disease Dataset

The dataset contains features such as age, sex, cholesterol, resting blood pressure, fasting glucose levels, resting electrocardiographic measurement, maximum heart rate achieved, Exercise induced angina, calcium and thalassemia.

```

RangeIndex: 303 entries, 0 to 302
Data columns (total 14 columns):
 #   Column      Non-Null Count  Dtype  
---  -
 0   age         303 non-null   int64  
 1   sex         303 non-null   int64  
 2   cp          303 non-null   int64  
 3   trestbps    303 non-null   int64  
 4   chol        303 non-null   int64  
 5   fbs         303 non-null   int64  
 6   restecg     303 non-null   int64  
 7   thalach     303 non-null   int64  
 8   exang       303 non-null   int64  
 9   oldpeak     303 non-null   float64 
10   slope       303 non-null   int64  
11   ca          303 non-null   int64  
12   thal        303 non-null   int64  
13   target      303 non-null   int64  
dtypes: float64(1), int64(13)
memory usage: 33.3 KB

```

5.1.4 Parkinson's Disease Dataset

The dataset was created by Max Little of the University of Oxford, in collaboration with the National Centre for Voice and Speech, Denver, Colorado, who recorded the speech signals. The original study published the feature extraction methods for general voice disorders. This dataset is composed of a range of biomedical voice measurements from 31 people, 23 with Parkinson's disease (PD). Each column in the table is a particular voice measure, and each row corresponds one of 195 voice recording from these individuals ("name" column). The main aim of the data is to discriminate healthy people from those with PD, according to "status" column which is set to 0 for healthy and 1 for PD.

columns	Description
name	ASCII subject name and recording number
MDVP:F0(Hz)	Average vocal fundamental frequency
MDVP:Fhi(Hz)	Maximum vocal fundamental frequency
MDVP:Flo(Hz)	Minimum vocal fundamental frequency
MDVP:Jitter(%),MDVP:Jitter(Abs),MDVP:RAP,MDVP:PPQ,Jitter:DDP	Several measures of variation in fundamental frequency
MDVP:Shimmer,MDVP:Shimmer(dB),Shimmer:APQ3,Shimmer:APQ5,MDVP:APQ,Shimmer:DDA	Several measures of variation in amplitude
NHR,HNR	Two measures of ratio of noise to tonal components in the voice
status	Health status of the subject (one) - Parkinson's, (zero) - healthy
RPDE,D2	Two nonlinear dynamical complexity measures
DFA	Signal fractal scaling exponent
spread1,spread2,PPE	Three nonlinear measures of fundamental frequency variation

Information about the dataset.

```
▶ RangeIndex: 195 entries, 0 to 194
Data columns (total 24 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   name                                  195 non-null    object
1   MDVP:Fo(Hz)                          195 non-null    float64
2   MDVP:Fhi(Hz)                         195 non-null    float64
3   MDVP:Flo(Hz)                         195 non-null    float64
4   MDVP:Jitter(%)                      195 non-null    float64
5   MDVP:Jitter(Abs)                    195 non-null    float64
6   MDVP:RAP                             195 non-null    float64
7   MDVP:PPQ                             195 non-null    float64
8   Jitter:DDP                           195 non-null    float64
9   MDVP:Shimmer                         195 non-null    float64
10  MDVP:Shimmer(dB)                    195 non-null    float64
11  Shimmer:APQ3                        195 non-null    float64
12  Shimmer:APQ5                        195 non-null    float64
13  MDVP:APQ                             195 non-null    float64
14  Shimmer:DDA                          195 non-null    float64
15  NHR                                  195 non-null    float64
16  HNR                                  195 non-null    float64
17  status                              195 non-null    int64
18  RPDE                                 195 non-null    float64
19  DFA                                  195 non-null    float64
20  spread1                             195 non-null    float64
21  spread2                             195 non-null    float64
22  D2                                   195 non-null    float64
23  PPE                                  195 non-null    float64
dtypes: float64(22), int64(1), object(1)
memory usage: 36.7+ KB
```

5.2 Feature Extraction

After loading the data, checking for missing values using `.isnull()` function. Then features like `patient_id` (if present) and other features that will not help in shaping the data are dropped. Splitting of features and target is done, the important features are kept and plotted along x axis, and the target values (1 for negative result, 0 for positive result) are plotted along y axis.

The data is divided into two parts. one part (80% of the original data) will be used for training the model, and rest (20% of the original data) will be used for testing purpose. The data is then fitted into the machine learning

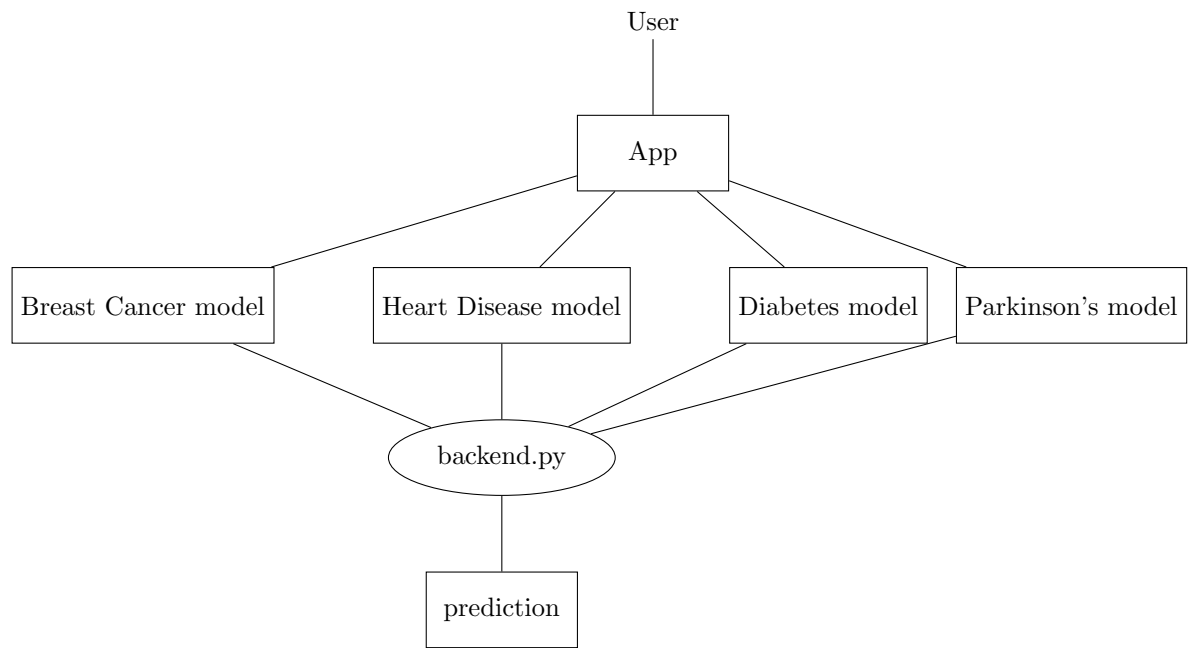
models. Accuracy tests of each model for their respective disease is evaluated and the model for both training and testing data. Model with the highest accuracy score is saved and will be used for prediction of that disease.

5.3 Frontend

A react.js file named 'App.jsx' is created which holds the path to other application for diseases. This file defines the App component for a React application. The App component serves as the main container for the user interface, including navigation and different disease prediction forms. The component imports several dependencies, including React for building the UI, various disease-specific components like Diabetes, Heart Disease, Breast Cancer, and Parkinson's, and a Sidebar component for navigation. It also imports a CSS file for styling. Inside the App component, a state variable called 'selectedOption' is used to track the currently selected disease prediction form. This state is initialized to display the "Diabetes Prediction" form by default. The 'setSelectedOption' function is used to update this state based on user interaction. 'Sidebar.jsx' is the main gateway to different disease pages. The component renders a Sidebar component, which provides navigation options. Based on the value of selectedOption, the component conditionally renders the corresponding disease prediction form. If "Diabetes Prediction" is selected, the Diabetes component is shown. Similarly, the Heart Disease, Breast Cancer, or Parkinson's components are rendered based on the user's selection. The App component is exported as the default export, making it available for use in other parts of the application. Overall, this file defines the primary structure and functionality for switching between different disease prediction forms in the application, handling user selection through the Sidebar, and managing the content displayed based on the user's choice.

5.4 Backend

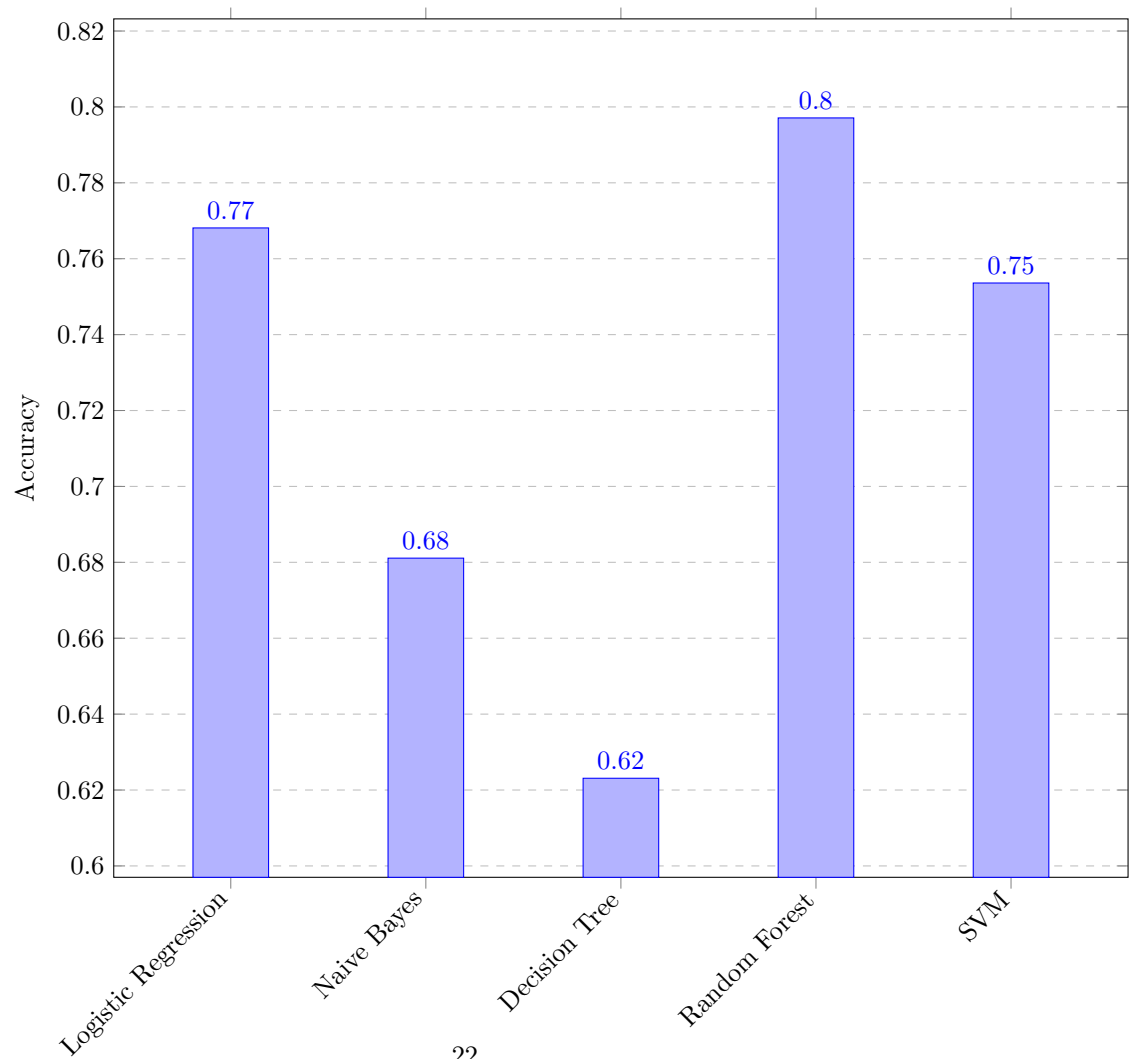
Flask is a lightweight Python web framework for building web applications. It provides essential tools for routing, request handling, and templating, allowing developers to create flexible and scalable applications. Its minimalist approach gives developers control over the project structure, making it suitable for small to medium-sized projects. In this project, the data provided by the user at the 'App.jsx' file is fetched. At the backend, all trained machine learning models are loaded using pickle or joblib, and they are provided with the data of the specific disease. The model then makes a prediction and a value (0/1) is generated. The python file then returns a result in a statement whether there is a probability of the disease from the given symptoms or not.



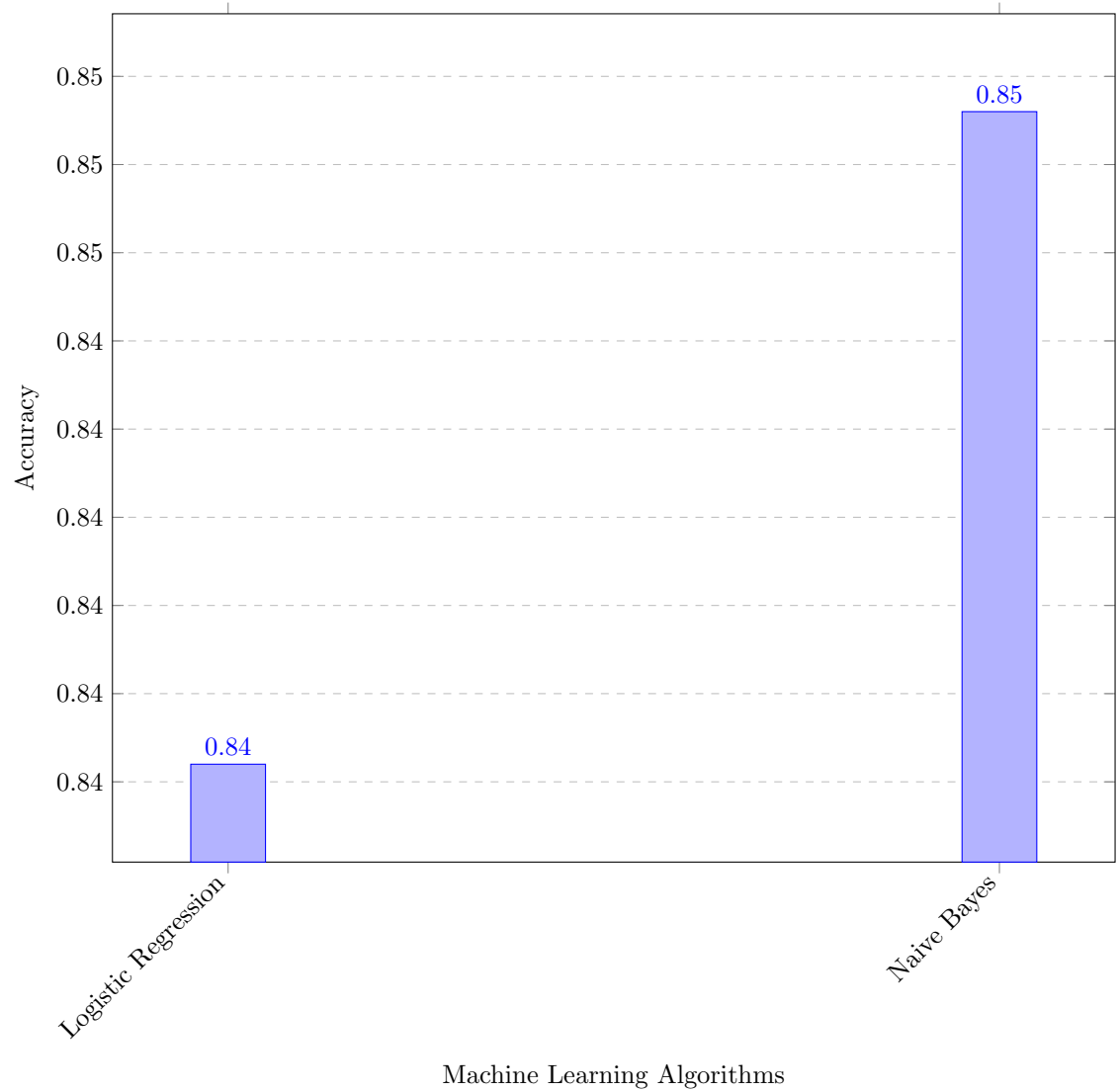
Result And Discussion

6.1 Accuracy Scores

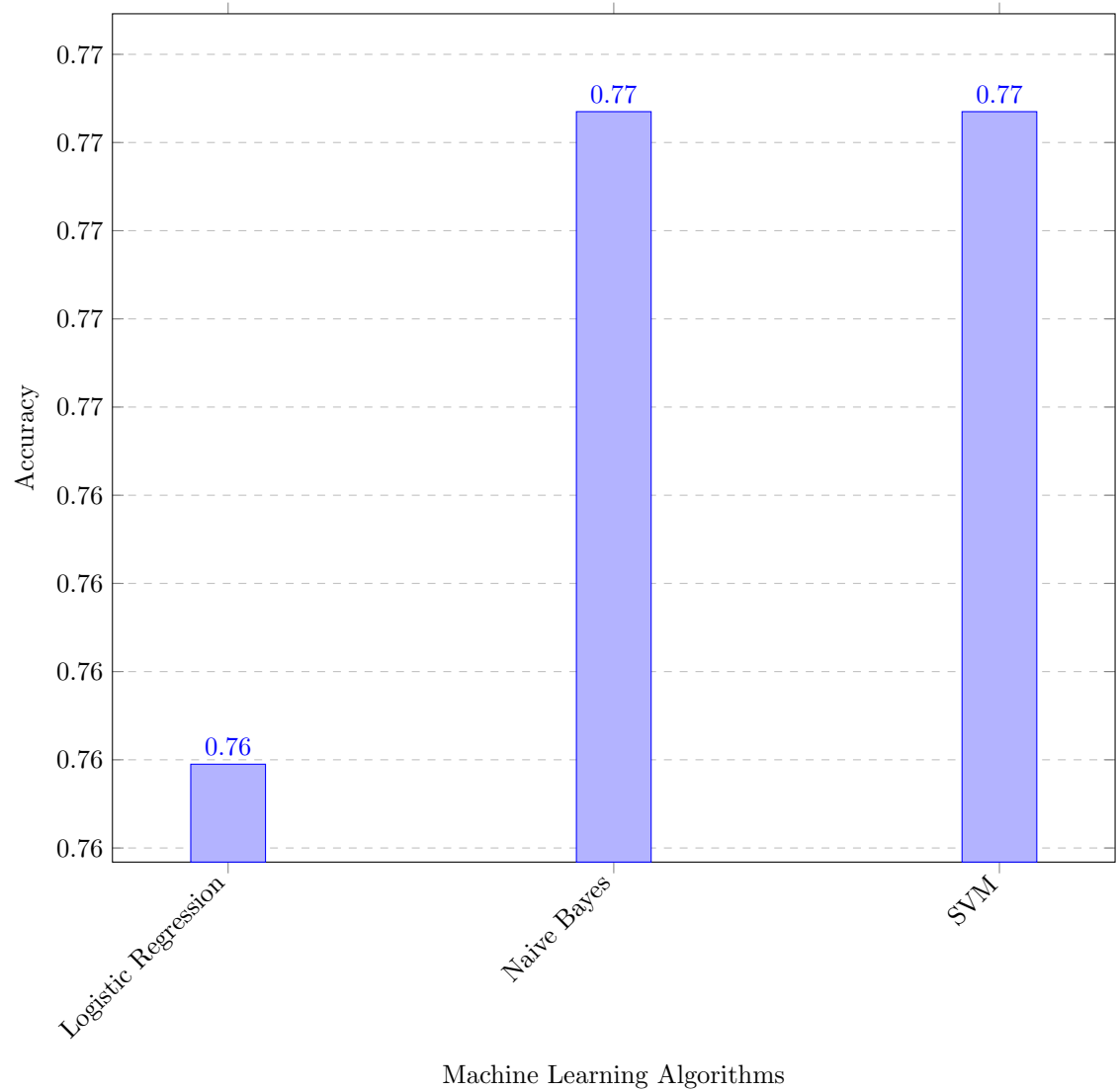
6.1.1 Breast Cancer Prediction



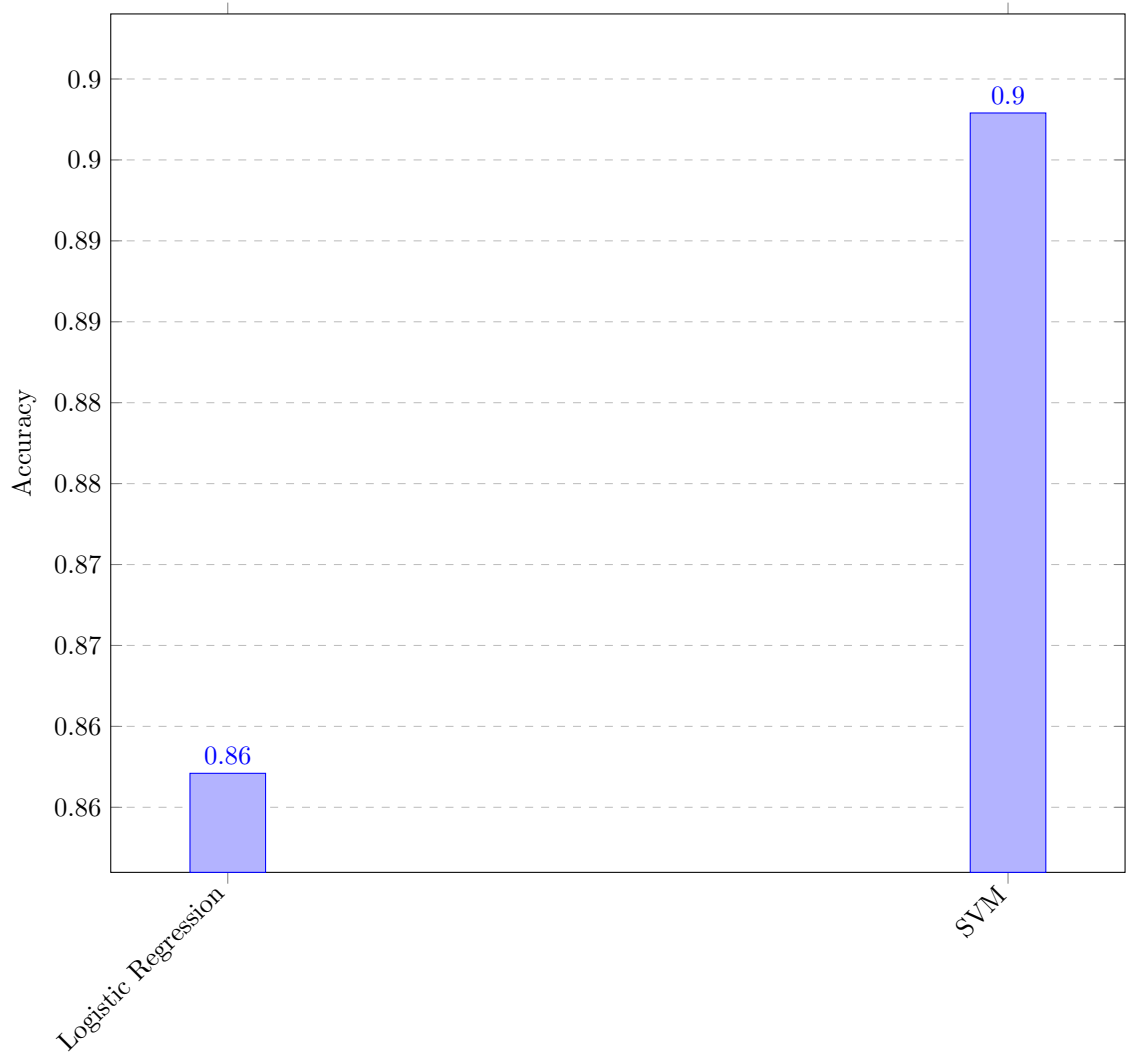
6.1.2 Heart Disease Prediction



6.1.3 Diabetes Prediction



6.1.4 Parkinson's Disease Prediction



Machine Learning Algorithms

The fields of disease diagnosis and prediction hold significant potential for transformation through ML techniques. Accuracy and correctness in diagnosis are most important aspects in effective treatment procedures for managing sickness. In our system, we employ the Random forest classifier for breast cancer, Naive Bayes algorithm for heart disease, SVM for diabetes and Parkinson's disease for prediction. Patients input values into the system to determine disease presence. The accuracy of predictions relies on the machine learning algorithm's ability to retrieve precise outputs.

Future Improvements

- Any inaccuracy in the model is a result of model underfitting. So gathering a sufficient data, and re-training the model with sufficient and relevant dataset is very important for improving accuracy.
- Extending the scope of the application to include a broader range of diseases. This involves collecting data and developing models for additional diseases, thus providing a more comprehensive diagnostic tool.
- Identifying and focusing on diseases that are prevalent or of significant concern, while also considering the development of models for less common diseases.
- Conducting a thorough analysis to identify the most relevant features for disease prediction. Implementing advanced feature selection and extraction techniques to enhance model performance.
- Collaborating with medical professionals to ensure that the features used in the models are clinically relevant and contribute to accurate predictions.
- Continuously benchmarking the performance of the models against existing ones to identify areas for improvement and ensure that the system offers better or competitive accuracy is necessary.
- The accuracy scores of the models are either a little better or almost exactly equal to existing models. So, there is a scope for identifying relevant machine learning models for predicting a specific disease.

Conclusion

The main objective of this research was to create a system that could reliably forecast several illnesses. The user is spared from having to visit multiple websites. Early diagnosis of diseases can both lengthen your life and spare you from financial hardship. To achieve the highest level of accuracy, we have used a variety of machine learning algorithms. In this study, we investigated the use of machine learning methods for the prognosis of various illnesses, with an emphasis on breast cancer, heart disease, diabetes, and Parkinson's disease. Using the machine learning model with highest accuracy with their respective disease, we created a framework for multiple disease prediction. To guarantee data quality, we gathered information from Kaggle.com and preprocessed it. We used SVM algorithm and obtained almost 90% accuracy for parkinson's disease prediction, and 77% accuracy for diabetes prediction. Similarly, we used Naive Bayes to predict Heart disease with an 85% accuracy rate. An accuracy of 80% was obtained when Breast cancer was predicted using Random Forest Classifier. The results of this study show how machine learning has the power to transform illness prediction and enhance patient outcomes. Various machine learning model's im-

plementation required handling and filtering. When a trained model is included into an application, it may forecast diseases in real-world situations, enabling researchers, healthcare providers, and individuals to make well-informed decisions about managing and assessing disease risk. Targeted illness management techniques, individualized treatment regimens, and early interventions may all be made easier with the help of machine learning models for accurate disease prediction. It can improve patient care, help healthcare providers make well-informed decisions, and optimize the distribution of resources within healthcare systems. It also has potential for population-level disease surveillance, which would allow for the early identification of disease outbreaks and the application of preventative measures. In summary, this study highlights the potential of machine learning in multi-disease prediction and advances the field of disease prediction using machine learning. We can get closer to developing more precise, timely, and individualized healthcare interventions by utilizing machine learning, which will ultimately enhance patient outcomes and create more effective healthcare systems.

Screenshots

8.1 Diabetes Prediction

The screenshot shows a web browser window with two tabs, both titled "Multiple Disease Prediction". The address bar shows "localhost:3000". On the left side of the browser, there is a sidebar with the text "Multiple Disease Prediction System" and a list of four prediction options: "Diabetes Prediction" (highlighted in red), "Heart Disease Prediction", "Breast Cancer Prediction", and "Parkinson's Disease Prediction". The main content area of the browser displays a "Diabetes Prediction" form. The form has a red header with the text "Diabetes Prediction". Below the header, there are five input fields, each preceded by a label: "Number of Pregnancies:", "Glucose:", "Blood Pressure:", "Skin Thickness:", and "Insulin:". The Windows taskbar is visible at the bottom of the screen, showing the Start button, a search bar, and several application icons. The system tray on the right shows the language set to "ENG IN", the date and time as "18:58 04-08-2024", and icons for network, volume, and battery.

Multiple Disease Prediction System

Diabetes Prediction

Heart Disease Prediction

Breast Cancer Prediction

Parkinson's Disease Prediction

Diabetes Prediction

Number of Pregnancies:

Glucose:

Blood Pressure:

Skin Thickness:

Insulin:

8.1.1 Inputs

Multiple Disease Prediction System

Diabetes Prediction

Heart Disease Prediction

Breast Cancer Prediction

Parkinson's Disease Prediction

Diabetes Prediction

Number of Pregnancies:

4

Glucose:

103

Blood Pressure:

60

Skin Thickness:

33

Insulin:

192

Skin Thickness:

33

Insulin:

192

BMI:

24

Diabetes Pedigree:

0.966

Age:

63

Submit

8.1.2 Output

The screenshot shows a web browser window with two tabs, both titled "Multiple Disease Prediction". The address bar shows "localhost:3000". The page has a sidebar on the left with the title "Multiple Disease Prediction System" and four buttons: "Diabetes Prediction" (highlighted in red), "Heart Disease Prediction", "Breast Cancer Prediction", and "Parkinson's Disease Prediction". The main content area contains several input fields with labels and a "Submit" button. The inputs are: "Skin thickness:" with value "33", "Insulin:" with value "192", "BMI:" with value "24", "Diabetes Pedigree:" with value "0.966", and "Age:" with value "63". Below the "Submit" button, a green box displays the prediction: "Prediction: This person is Diabetic." The Windows taskbar at the bottom shows the search bar, task view button, and several application icons. The system tray on the right shows the language set to "ENG IN", network and volume icons, and the date and time "18:59 04-08-2024".

Multiple Disease Prediction System

Diabetes Prediction

Heart Disease Prediction

Breast Cancer Prediction

Parkinson's Disease Prediction

Skin thickness:

33

Insulin:

192

BMI:

24

Diabetes Pedigree:

0.966

Age:

63

Submit

Prediction: This person is Diabetic.

8.2 Heart Prediction

The screenshot shows a web browser window with two tabs, both titled "Multiple Disease Prediction". The address bar shows "localhost:3000". The web application has a sidebar on the left with the title "Multiple Disease Prediction System". Under this title, there are four menu items: "Diabetes Prediction", "Heart Disease Prediction" (which is highlighted with a red background), "Breast Cancer Prediction", and "Parkinson's Disease Prediction". The main content area on the right has a red header with the text "Heart Disease Prediction". Below this header, there are five input fields, each preceded by a label: "Age:", "Sex (0=Male, 1=Female):", "Constrictive Pericarditis:", "Test BPS:", and "Cholesterol:". The Windows taskbar is visible at the bottom of the screen, showing the Start button, a search bar, and several application icons. The system tray on the right shows the date and time as "19:13 04-08-2024".

Multiple Disease Prediction System

- Diabetes Prediction
- Heart Disease Prediction
- Breast Cancer Prediction
- Parkinson's Disease Prediction

Heart Disease Prediction

Age:

Sex (0=Male, 1=Female):

Constrictive Pericarditis:

Test BPS:

Cholesterol:

8.2.1 Inputs

The screenshot displays a web browser window with two tabs, both titled "Multiple Disease Prediction". The address bar shows "localhost:3000". The page features a sidebar on the left with the heading "Multiple Disease Prediction System" and a list of prediction options: "Diabetes Prediction", "Heart Disease Prediction" (highlighted in red), "Breast Cancer Prediction", and "Parkinson's Disease Prediction". The main content area is titled "Heart Disease Prediction" in a red box. It contains several input fields with the following labels and values:

- Age:** 41
- Sex (0=Male, 1=Female):** 1
- Constrictive Pericarditis:** 0
- Test BPS:** 110
- Cholesterol:** 172
- Fasting Blood Sugar (0=No, 1=yes):** 0
- Rest CG (0=Normal, 1=Abnormal):** 0
- Thalach (Maximum heart rate achieved. 0=No, 1=Yes):** 158
- Exercise induced Angina (0=No, 1=Yes):** 0
- Old Peak:** 0

The Windows taskbar at the bottom shows the search bar, task view button, and several application icons. The system tray on the right indicates the language is "ENG IN", and the date and time are "04-08-2024" and "19:14".

Multiple Disease Prediction System

Diabetes Prediction

Heart Disease Prediction

Breast Cancer Prediction

Parkinson's Disease Prediction

Exercise induced Angina (0=No, 1=Yes):

0

Old Peak:

0

Slope:

2

CAD:

0

Thalassemia:

3

Submit

8.2.2 Output

Multiple Disease Prediction System

Diabetes Prediction

Heart Disease Prediction

Breast Cancer Prediction

Parkinson's Disease Prediction

Exercise induced Angina (0=No, 1=Yes):

0

Old Peak:

0

Slope:

2

CAD:

0

Thalassemia:

3

Submit

Prediction: This person does not have Heart Disease.

8.3 Breast Cancer Prediction

The screenshot shows a web browser window with two tabs, both titled "Multiple Disease Prediction". The address bar shows "localhost:3000". On the left is a sidebar menu with the following items: "Multiple Disease Prediction System", "Diabetes Prediction", "Heart Disease Prediction", "Breast Cancer Prediction" (highlighted in red), and "Parkinson's Disease Prediction". The main content area has a red header "Breast Cancer Prediction". Below it are five input fields with the following labels: "Age", "Menopausal status (0= premenopausal, 1= postmenopausal)", "Tumor Size:", "Tumor Grade", and "Number Of Positive Lymph Nodes:". The Windows taskbar at the bottom shows the Start button, a search bar, and several application icons. The system tray on the right shows the language "ENG IN", network and volume icons, and the date and time "19:15 04-08-2024".

Multiple Disease Prediction System

Diabetes Prediction

Heart Disease Prediction

Breast Cancer Prediction

Parkinson's Disease Prediction

Breast Cancer Prediction

Age

Menopausal status (0= premenopausal, 1= postmenopausal):

Tumor Size:

Tumor Grade

Number Of Positive Lymph Nodes:

8.3.1 Inputs

Multiple Disease Prediction System

- Diabetes Prediction
- Heart Disease Prediction
- Breast Cancer Prediction**
- Parkinson's Disease Prediction

Breast Cancer Prediction

Age

Menopausal status (0= premenopausal, 1= postmenopausal):

Tumor Size:

Tumor Grade

Number Of Positive Lymph Nodes:

Number Of Positive Lymph Nodes:

Progesterone Receptors (fmol/l):

Estrogen Receptors (fmol/l):

Hormonal Therapy (0=no, 1=yes):

Recurrence Free Survival Time;days to first of recurrence, death or last follow-up:

Submit

8.3.2 Output

The screenshot shows a web browser window with two tabs, both titled "Multiple Disease Prediction". The address bar shows "localhost:3000". The web application has a sidebar on the left with the following menu items: "Multiple Disease Prediction System", "Diabetes Prediction", "Heart Disease Prediction", "Breast Cancer Prediction" (highlighted in red), and "Parkinson's Disease Prediction". The main content area displays the following input fields and their values:

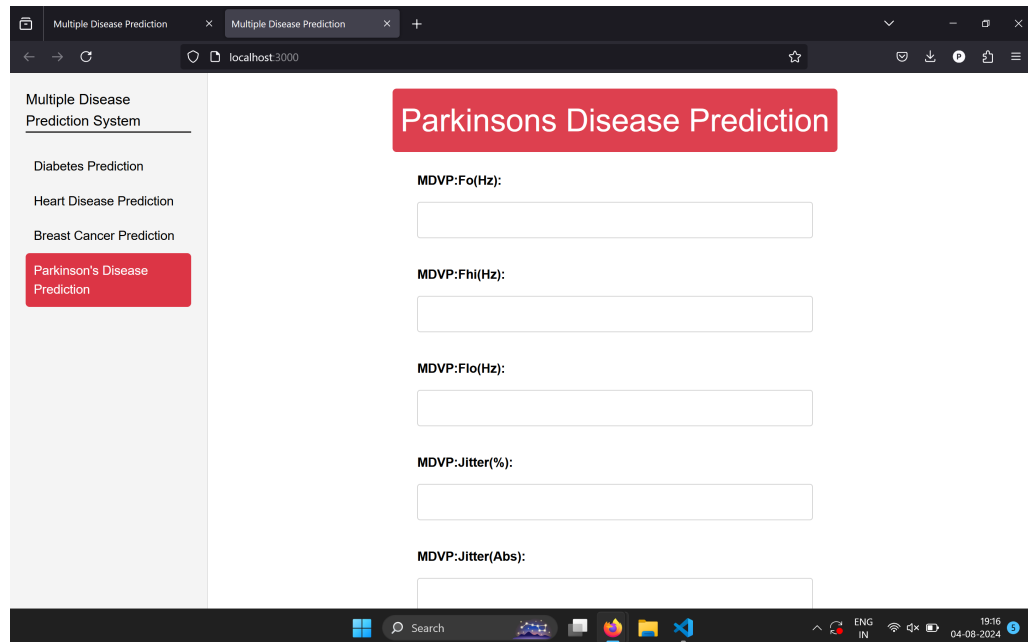
- 4
- Progesterone Receptors (fmol/l): 10
- Estrogen Receptors (fmol/l): 10
- Hormonal Therapy (0=no, 1=yes): 0
- Recurrence Free Survival Time;days to first of recurrence, death or last follow-up: 2456

Below the input fields is a "Submit" button. The output of the prediction is displayed in a green box:

The prediction is: Alive without Recurrence (Breast cancer recurrence is when the symptoms return after initial treatment)

The Windows taskbar at the bottom shows the search bar, task view button, and several application icons. The system clock indicates the time is 19:16 on 04-08-2024.

8.4 Parkinson's Disease Prediction



Multiple Disease Prediction System

- Diabetes Prediction
- Heart Disease Prediction
- Breast Cancer Prediction
- Parkinson's Disease Prediction**

Parkinsons Disease Prediction

MDVP:F0(Hz):

MDVP:F1(Hz):

MDVP:F1o(Hz):

MDVP:Jitter(%):

MDVP:Jitter(Abs):

8.4.1 Inputs

The screenshot displays a web browser window with two tabs, both titled "Multiple Disease Prediction". The address bar shows "localhost:3000". On the left side of the browser, there is a sidebar menu for the "Multiple Disease Prediction System". The menu items are: "Diabetes Prediction", "Heart Disease Prediction", "Breast Cancer Prediction", and "Parkinson's Disease Prediction". The "Parkinson's Disease Prediction" item is highlighted with a red background. The main content area of the browser shows the "Parkinsons Disease Prediction" page. It features a series of input fields for different MDVP features. The visible inputs and their values are: MDVP:Fo(Hz): 180.97800, MDVP:Fhi(Hz): 200.12500, MDVP:Flo(Hz): 155.495, MDVP:Jitter(%): 0.00406, MDVP:Jitter(Abs): 0.00002, MDVP:Jitter(Abs): 0.00002, MDVP:RAP: 0.00220, MDVP:PPQ: 0.00244, Jitter:DDP: 0.00659, MDVP:Shimmer: 0.03852, and MDVP:Shimmer(dB): 0.33100. The browser's taskbar at the bottom shows the Windows logo, a search bar, and several application icons. The system tray on the right indicates the language is "ENG IN", the date is "04-08-2024", and the time is "19:18".

Multiple Disease Prediction System

- Diabetes Prediction
- Heart Disease Prediction
- Breast Cancer Prediction
- Parkinson's Disease Prediction

Parkinsons Disease Prediction

MDVP:Fo(Hz):
180.97800

MDVP:Fhi(Hz):
200.12500

MDVP:Flo(Hz):
155.495

MDVP:Jitter(%):
0.00406

MDVP:Jitter(Abs):
0.00002

MDVP:Jitter(Abs):
0.00002

MDVP:RAP:
0.00220

MDVP:PPQ :
0.00244

Jitter:DDP:
0.00659

MDVP:Shimmer:
0.03852

MDVP:Shimmer(dB):
0.33100

Multiple Disease Prediction System

Diabetes Prediction

Heart Disease Prediction

Breast Cancer Prediction

Parkinson's Disease Prediction

MDVP:Shimmer(dB):

0.33100

Shimmer:APQ3:

0.02107

Shimmer:APQ5:

0.02493

MDVP:APQ:

0.02877

Shimmer:DDA:

0.06321

NHR:

0.0278

Multiple Disease Prediction System

Diabetes Prediction

Heart Disease Prediction

Breast Cancer Prediction

Parkinson's Disease Prediction

NHR:

0.0278

HNR:

16.176

RPDE:

0.5835

DFA:

0.7277

Spread1:

-5.6578

Spread2:

0.3159

Multiple Disease Prediction System

Diabetes Prediction

Heart Disease Prediction

Breast Cancer Prediction

Parkinson's Disease Prediction

DFA:

0.7277

Spread1:

-5.6578

Spread2:

0.3159

D2:

3.098

PPE:

0.2004

Submit

8.4.2 Output

Multiple Disease Prediction System

Diabetes Prediction

Heart Disease Prediction

Breast Cancer Prediction

Parkinson's Disease Prediction

DFA:

0.7277

Spread1:

-5.6578

Spread2:

0.3159

D2:

3.098

PPE:

0.2004

Submit

Prediction: This person does not have Parkinson's Disease.